Q.1 Check if the following limit exists. if yes find the limiting value

Write step by step on how you'd prove/ disprove for limit existance

Q.1 i) $\lim_{(x,y)\to(0,0)} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}}$ — 2

Q.2 ii) $\lim_{(x,y)\to(0,0)} \frac{x^2 y}{x^4 + y^2}$ — 2

Ans.

i) a. along $x = 0$, limit becomes

$$\lim_{(y\to 0)} \frac{0 - 0}{0 - \sqrt{y}} = 0$$

b. along $y = 0$, limit becomes

$$\lim_{(x\to 0)} \frac{x^2 - 0}{\sqrt{x} - 0} = 0$$

c. along $y = kx$, function becomes

$$\lim_{x\to 0} \frac{x^2 - kx^2}{\sqrt{x} - \sqrt{x}\sqrt{k}}$$

$$x^2 (1 - k)$$

$$\lim_{x \to 0} \frac{}{\sqrt{x}\,(1-\sqrt{k})}$$

$$\lim_{x \to 0} x^{3/2} \frac{(1-\sqrt{k})(1+\sqrt{k})}{(1-\sqrt{k})}$$

$$\lim_{x \to 0} x^{3/2} (1+\sqrt{k}) = 0$$

powers of x & y in numerator $>$ powers of x & y in denominator

Hence, the limiting value will approach to 0.

ii) along $x = 0$, limit becomes

$$\lim_{(y \to 0)} \frac{0 \times y}{0 + y^2}$$

$$\lim_{y \to 0} \frac{0}{y^2} = 0$$

along $y = 0$ it becomes

$$\lim_{x \to 0} \frac{0}{x^4} = 0$$

along $y = kx$

$$\lim_{x \to 0} \frac{x^2 \times kx}{x^4 + k^2 x^2}$$

$$\lim \frac{kx}{\quad + k^2} = \frac{0}{k^2} = 0$$

$x \to 0 \quad x^2 + k$

for $k \neq 0$

since power of x & y in numerator is less than power in denominator, we need to check along other curves as wel

d. check along $y = x^2$

$$\lim_{x \to 0} \frac{x^2 \times x^2}{x^4 + (x^2)^2}$$

$$\lim_{x \to 0} \frac{x^4}{x^4(2)} \to \left(\tfrac{1}{2}\right)$$

limiting value does not approach to a single value so Limit does not exist.

Q.3 Find unit vector along the direction of gradient for this function at $x = 1$, $y = 1$

$$f(x,y) = x^2 y^2 + xy + y$$

Ans.

$$\frac{\partial f}{\partial x} = 2xy^2 + y$$

$$\frac{\partial f}{\partial y} = 2yx^2 + x + 1$$

$$\left.\frac{\partial f}{\partial x}\right|_{(1,1)} = 3 \quad , \quad \left.\frac{\partial f}{\partial y}\right|_{(1,1)} = 4$$

$$\nabla f = 3\hat{i} + 4\hat{j}$$

unit vector $\left\langle \frac{3}{5}, \frac{4}{5} \right\rangle$ — (2)

Q find all critical points for the given function and decide whether each of these point is minima or maxima or a saddle or none of these

Note: The functions are continuous in the entire domain

Q4 i) $f(x,y) = 2xy + 3x + 4y$ — (1)

Q5 ii) $f(x,y) = x^2 + y - e^y$ — (1)

Ans i) $\frac{\partial f}{\partial x} = 2y + 3 \quad , \quad \frac{\partial f}{\partial y} = 2x + 4$

$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0$ gives $y = \frac{-3}{2}, x = -2$

$$\frac{\partial^2 f}{\partial x^2} = 0 \quad , \quad \frac{\partial^2 f}{\partial x \partial y} = 2 \quad , \quad \frac{\partial^2 f}{\partial y \partial x} = 2 \quad , \quad \frac{\partial^2 f}{\partial y^2} = 0$$

$$D = f_{xx} f_{yy} - f_{xy}^2$$

$$= -4 < 0 \quad \text{Saddle}$$

Critical Point $\left(-2, -\frac{3}{2}, -6\right)$ is a Saddle point

ii) $\frac{\partial f}{\partial x} = 2x$, $\frac{\partial f}{\partial y} = 1 - e^y$

$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0 \Rightarrow x = 0, \; y = 0$

critical point $(0, 0, -1)$

$f_{xx} = 2$, $f_{yy} = -e^y$, $f_{xy} = 0$

$f_{yy}\big|_{(0,0,-1)} = -1$

$D = f_{xx} f_{yy} - f_{xy}^2$

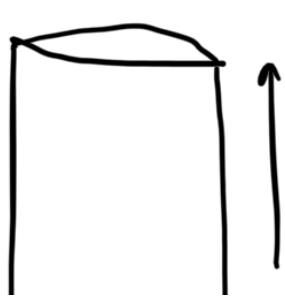
$= (2)(-1) - 0 = -2 < 0$

Saddle

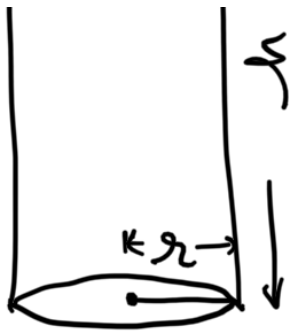
Q.6 Find the maximum volume of a cylindrical soda can such that the sum of its height & circumference is 120 cm. — (8)

Ans. circumference of a cylinder $= 2\pi r$,

where r is radius of the circular cross-section



Volume of cylinder $= \pi r^2 h$



$$f(r,h) = \pi r^2 h$$

$$g(r,h) = 2\pi r + h$$

$$g(r,h) = k \quad , \quad k = 120 \text{ cm}$$

maximize $f(r,h)$ with given boundary of $g(r,h)$

<u>Method 1:</u>

parametrize $g(r,h)$

$$h(t) = t \quad , \quad r(t) = \frac{k-t}{2\pi}$$

$$g(t) = 2\pi\left(\frac{k-t}{2\pi}\right) + t = k$$

$$\frac{df}{dt} = \frac{\partial f}{\partial r}\cdot\frac{dr}{dt} + \frac{\partial f}{\partial h}\cdot\frac{dh}{dt}$$

$$= 2\pi r h\cdot\left(\frac{-1}{2\pi}\right) + \pi r^2(1)$$

$$= -rh + \pi r^2 = 0$$

$$r(\pi r - h) = 0$$

either $r=0$ or $r = \frac{h}{\pi}$ are our critical points to look at

$r=0$ will make the volume $=0$, so it won't give us the max volume so

$r = \frac{h}{\pi}$ should be the critical point for max volume

Now, $2\pi r + h = 120$ cm

$2h + h = 120$ cm

$h = 40$ cm

$r = \frac{40}{\pi}$ cm

max Volume $V = \pi \left(\frac{40}{\pi}\right)^2 \times 40$

$= \frac{40^3}{\pi}$ cm^3

$$\boxed{V_{max} = 20.38 \times 10^3 \text{ cm}^3}$$

alternatively we could check $\frac{d^2f}{dt^2}$ to prove it is a maxima, that part is optional

<u>Method (2)</u>:

$\frac{\partial f}{\partial h} = \pi r^2$, $\frac{\partial f}{\partial r} = 2\pi r h$

$\frac{\partial g}{\partial h} = 1$, $\frac{\partial g}{\partial r} = 2\pi$

$\nabla f = \lambda \nabla g$

$\langle \pi r^2, 2\pi r h \rangle = \lambda \langle 1, 2\pi \rangle$

$$\pi r^2 = \lambda$$

$$2\pi r h = 2\pi \lambda \Rightarrow \lambda = rh$$

gives $\pi r^2 = rh$

$$\frac{r}{h} = \frac{1}{\pi} \Rightarrow r = \frac{h}{\pi}$$

rest same as method 1

Q.7 Figure out which pair of vectors are orthogonal in the following:

[orthogonal means, they make 90° angle with each other]

$\vec{u} = \langle 3, 7, -2 \rangle$

$\vec{v} = \langle 5, -3, -3 \rangle$

$\vec{w} = \langle 0, 1, -1 \rangle$ — (2)

Ans. $\vec{u} \cdot \vec{v} = 15 - 21 + 6 = 0$ ✓

$\vec{u} \cdot \vec{w} = 0 + 7 + 2 = 9$ ✗

$\vec{v} \cdot \vec{w} = 0 - 3 + 3 = 0$ ✓

$\vec{u}, \vec{v}$ & $\vec{v}, \vec{w}$ are orthogonal

Q. write derivatives wrt x for the following

functions

i) $f(x) = \sqrt{x}\,\sin(x)$ — 1

ii) $f(x) = e^{-x} + \cos(\sin(x))$ — 1

iii) $f(x) = 1/x - x^3 + 2$ — 1

Ans. i) $f'(x) = \sqrt{x}\,.\cos x + \frac{1}{2\sqrt{x}}\sin(x)$

ii) $f'(x) = -e^{-x} + (-\sin(\sin x)).\cos x$

iii) $f'(x) = -\frac{1}{x^2} - 3x^2$

Q. Draw $f''(x)$, $f'(x)$ and $f(x)$, paying special attention to roots, tangents, inflection points and concavity/convexity

$$f(x) = x^3 - 6x^2 + 11x - 6$$ — 3

Ans.

$$f'(x) = 3x^2 - 12x + 11$$

$$f''(x) = 6x - 12$$

Inflection point $\Rightarrow x = 2$ $(f''(x) = 0)$

$x < 2$, function is concave

$x > 2$, function is convex

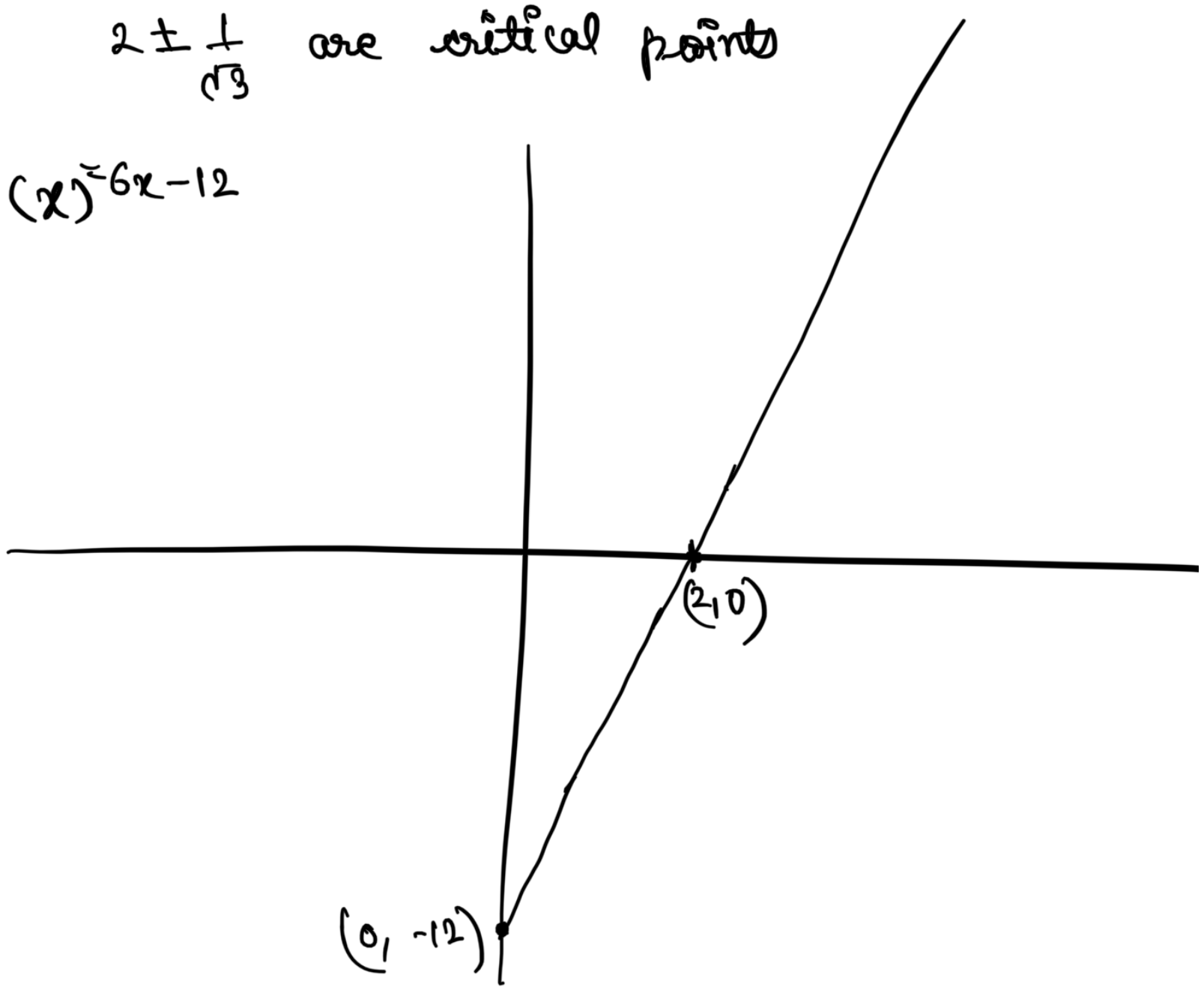
$f'(x) = 0$ gives

$$= \frac{-(-12) \pm \sqrt{144 - 4 \times 3 \times 11}}{2 \times 3}$$

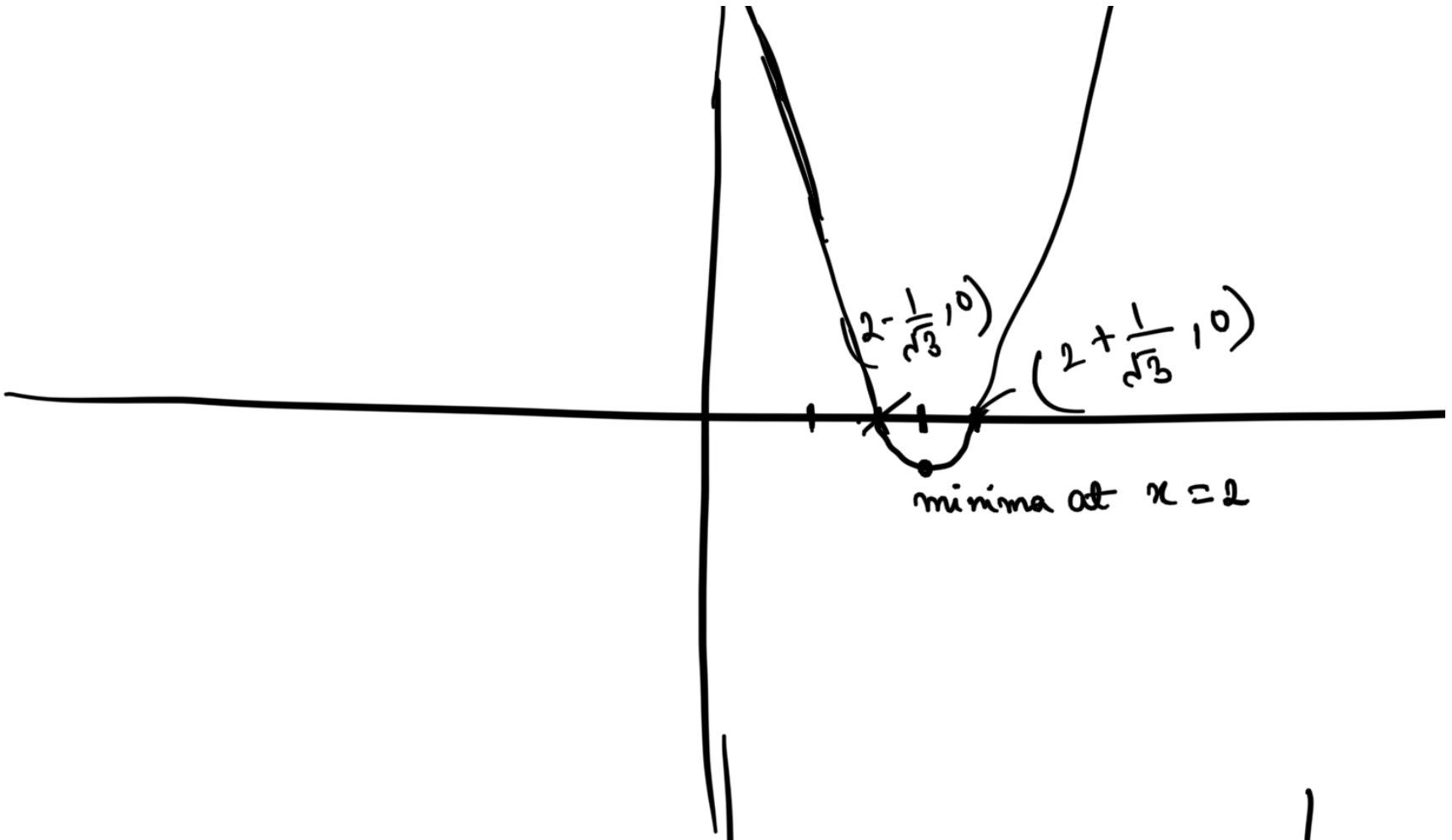
$$= \frac{12 \pm \sqrt{12}}{6}$$

$2 \pm \frac{1}{\sqrt{3}}$ are critical points

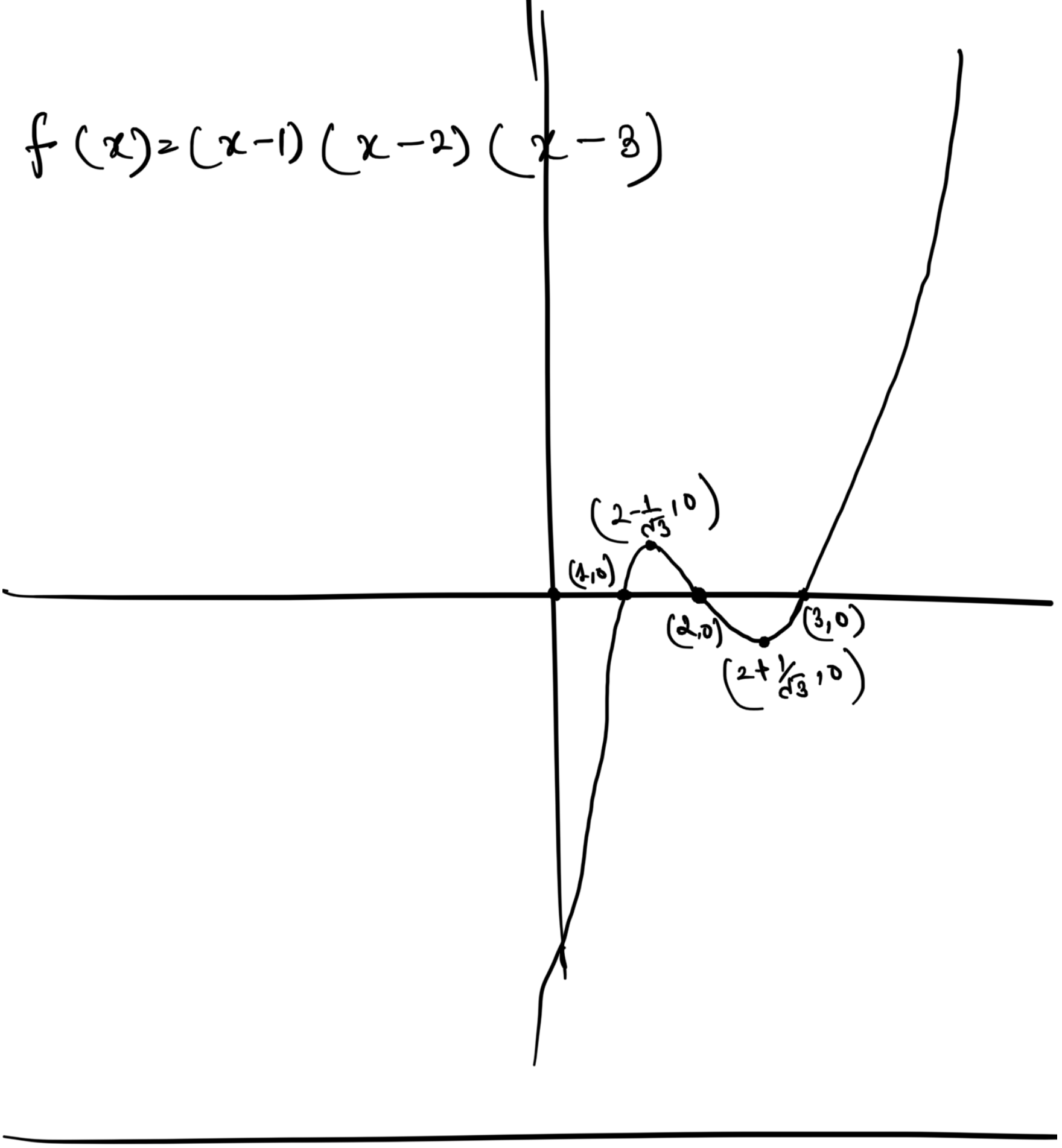
$f''(x) = 6x - 12$



$f'(x) = 3x^2 - 12x + 11$


$(2-\frac{1}{\sqrt{3}},0)$
$(2+\frac{1}{\sqrt{3}},0)$
minima at $x=2$

$f(x)=(x-1)(x-2)(x-3)$


$(2-\frac{1}{\sqrt{3}},0)$
$(1,0)$
$(2,0)$
$(3,0)$
$(2+\frac{1}{\sqrt{3}},0)$